

Cysteine Prediction in Pea Cultivars Using Raman Spectroscopy and Machine Learning

Literature Review

Submitted By:

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1. Pre-Processing Ensemble Modeling Based on Faster Covariate Selection Calibration for Near-Infrared Spectroscopy

Authors: Yonghong Wu, Yukun Zhou, Xiaojing Chen , Zhonghao Xie , Shujat Ali Leiming Yuan , Wen Shi , Xin Wang and Lechao Zhang

Overview:

Proposed PFCOVSC, a preprocessing ensemble strategy for near-infrared spectroscopy that addresses a fundamental limitation of conventional approaches, the need to select a single optimal preprocessing method. Rather than choosing one method, PFCOVSC fuses ten differently preprocessed versions of the same spectral dataset into a single matrix, including raw data, SNV, MSC, Savitzky-Golay smoothing, first and second derivatives, and their combinations. A fast covariate selection algorithm (fCovsel) then extracts the most informative variables across all preprocessing blocks simultaneously.

Dataset: The method was evaluated on three public NIR datasets — wheat protein (415 calibration, 108 test samples), meat fat content (172 calibration, 43 test), and tablet API content (210 training, 100 test). PFCOVSC reduced RMSEP by up to 49% on the meat dataset compared to the best single preprocessing method.

Crucially, it outperformed two established multi-block strategies, SPORT and PROSAC, across all three datasets while completing the entire optimization in under 0.5 seconds. PFCOVSC suggests that systematically combining multiple preprocessing methods and selecting variables across all of them could extract complementary spectral information that no single method captures, potentially improving LOCO generalization on the pea cysteine dataset.

2. Deep Learning in Spectral Analysis: Modeling and imaging

Authors: Xuyang Liu, Hongle An, Wensheng Cai , Xueguang Shao

Overview:

Liu et al. provide a comprehensive review of deep learning methods applied to spectral analysis, covering convolutional neural networks (CNNs), recurrent neural networks (RNNs), autoencoders, generative adversarial networks (GANs), and transfer learning. The review specifically addresses the two main criticisms of deep learning for spectroscopy: the requirement for large labeled datasets and the lack of interpretability.

For small datasets, the authors highlight three solutions: transfer learning (pretraining on large spectral datasets and fine-tuning on small ones), data augmentation (generating synthetic spectra by linearly combining existing ones), and variational autoencoders (VAEs) for unsupervised feature extraction. The review demonstrates that CNNs consistently outperform traditional methods

including PLSR and PCR on both large and small spectral datasets, and that they rely less on preprocessing than linear models because convolutional layers act as adaptive filters.

3. XGBoost algorithm assisted multi-component quantitative analysis with Raman spectroscopy

Authors: Qiaoyun Wang, Yinji Chen, Ziheng Zhu, Chongyue Yan, Peng Shan, Xin Zou, Yongqing Fu and Shuyu Wang

Overview:

Proposed a use of XGBoost not as a predictive model but as a preprocessing step for Raman spectral data. The XGBoost preprocessing method transforms raw spectral intensities into binary one-hot encoded feature vectors derived from decision tree leaf node assignments. These transformed features are then passed to either a linear regression (X-LR) or multilayer perceptron (X-MLP) model for concentration prediction.

The study used 40 Raman spectra of glucose, glycerol, and ethanol mixtures collected using a Fourier transform Raman spectrometer. XGBoost preprocessing was compared against eight conventional methods including SNV, MSC, Savitzky-Golay smoothing, first and second derivatives, wavelet transform, moving average, and mean centralization. The X-LR model achieved $R^2 = 0.9886$ for glucose prediction, outperforming all conventional preprocessing methods for both LR and MLP models.

4. Gaussian process regression for prediction and confidence analysis of fruit traits by near-infrared spectroscopy

Authors: Xiaojing Chen, Jianxia Xue, Xiao Chen, Xinyu Zhao, Shujat Ali, and Guangzao Huang

Overview:

Investigates Gaussian Process Regression (GPR) for predicting dry matter content in mango using NIR spectroscopy, with a specific focus on handling out-of-distribution (OOD) samples — a problem directly analogous to our LOCO evaluation where the test cultivar is entirely absent from training.

The dataset comprised 11,691 NIR spectra where 9,711 hard green mangoes used as in-distribution samples and 1,980 from ripe mangoes used as OOD samples. GPR was compared against PLSR and soft independent modeling of class analogy (SIMCA).

For in-distribution samples, GPR achieved RMSE=0.6908, outperforming PLSR (RMSE=0.8037).

The key advantage of GPR is that it produces both a predicted value and a variance estimate; larger variance indicates lower prediction confidence. Using the GPR variance as a threshold to filter unreliable predictions reduced RMSE on OOD samples by approximately 62%.

5. *A general deep learning model for predicting and classifying pea protein content via visible and near-infrared spectroscopy*

Authors: Tianpu Xiao, Dongxing Zhang, Chunji Xie, Tao Cui, Li Yang, Xiantao He , Kailiang Zhang, Liangju Wang , Hongsheng Li, Jiaqi Dong

Overview:

Xiao et al. developed a general deep learning model known as PeaNet Model for predicting and classifying pea protein content using visible and near-infrared spectroscopy, making this the most directly relevant paper to our study as it uses the same crop species. The model was designed to generalise across different pea varieties and growing conditions the same cross-cultivar generalisation challenge we face with cysteine prediction.

The study used Vis-NIR spectroscopy on pea samples from multiple varieties and proposed a dual-output deep learning architecture capable of simultaneously predicting protein concentration and classifying the cultivar. This joint learning approach allows the model to use cultivar identity as an auxiliary signal during training, potentially improving concentration prediction by learning variety-specific spectral patterns.

6. *Prediction of protein content in paddy rice (*Oryza sativa* L.) combining near-infrared spectroscopy and deep-learning algorithm*

Authors: Ha-Eun Yang¹, Nam-Wook Kim, Hong-Gu Lee, Min-Jee Kim, Wan-Gyu Sang, Changju Yang, Changyeun Mo

Overview:

Yang et al. (2024) investigated the use of near-infrared spectroscopy (NIRS) combined with machine learning and deep learning techniques for the non-destructive prediction of protein content in paddy rice and brown rice. The study compared three modelling approaches: Partial Least Squares Regression (PLSR), Support Vector Regression (SVR), and Deep Neural Network (DNN). A total of 1,800 paddy rice spectra and 1,200 brown rice spectra were collected in the wavelength range of 950–2200 nm, and models were developed using various spectral preprocessing methods.

The DNN model achieved the highest prediction accuracy among all tested approaches. The optimal DNN model, developed using first-order derivative preprocessing, produced a coefficient of determination for prediction (R_p^2) of 0.972 and an RMSEP of 0.048 g/100 g for paddy rice. For brown rice, the optimal DNN model achieved an R_p^2 of 0.987 and an RMSEP of 0.033 g/100 g. In comparison, both the PLSR and SVR models showed lower predictive performance.